New L J Institute of Engineering & Technology

SEM: 1st – CSE/CSE (AIML)/DS

Sub: Mathematics-I

Sub Code: BE01R00041

Module 1: Basic Calculus

Weightage: 20%

Topics: Applications of definite integral.

❖ Chap : Applications of definite integral

❖ Volume using Cross – Sections :

➢ This chapter deal with some important applications of definite integrals like Volume using cross sections and area of surface of revolution,

- In this section, we will study the method of slicing to find the volume of solids whose cross-sections are plane regions. This method is based on the formula

$$\text{Volume} = area \times height$$

- The volume of a solid of known integral cross- sectional area $A(x)$ from $x = a$ to $x = b$ is given by

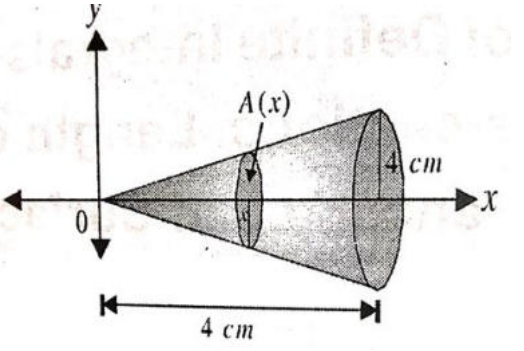
$$V = \int_a^b A(x)\, dx$$

- The volume of a solid of known integral cross- sectional area $A(y)$ from $y = c$ to $y = d$ is given by

$$V = \int_c^d A(y)\, dy$$

Example 1: Find the volume of a cone with height 4 cm and radius of base 4 cm. Use the method of slicing.

Solution:



Let us consider the cone is centred at the origin with its axis as the X- axis

From the figure the Cross-sectional area at x is

$$A(x) = \pi x^2$$

∴ The required volume is

$$V = \int_a^b A(x)\, dx$$

$$= \int_0^4 \pi x^2\, dx$$

$$= \pi \left(\frac{x^3}{3}\right)_0^4$$

$$= \pi \left(\frac{64}{3} - 0\right)$$

$$= \frac{64\pi}{3}\ cm^3$$

Example 2: Find the volume of the solid generated by rotating the plane region bounded by the curves $y = \frac{1}{x}$, x = 1 and x = 3 about the X-axis.

Solution: The volume is the generated by rotating the dotted region about the x- axis

The cross-section PQ of the generated volume is a circle of radius y perpendicular to x- axis

Area of Cross- section,

$$A = \pi y^2$$

$$= \pi \cdot \frac{1}{x^2}$$

The required volume is

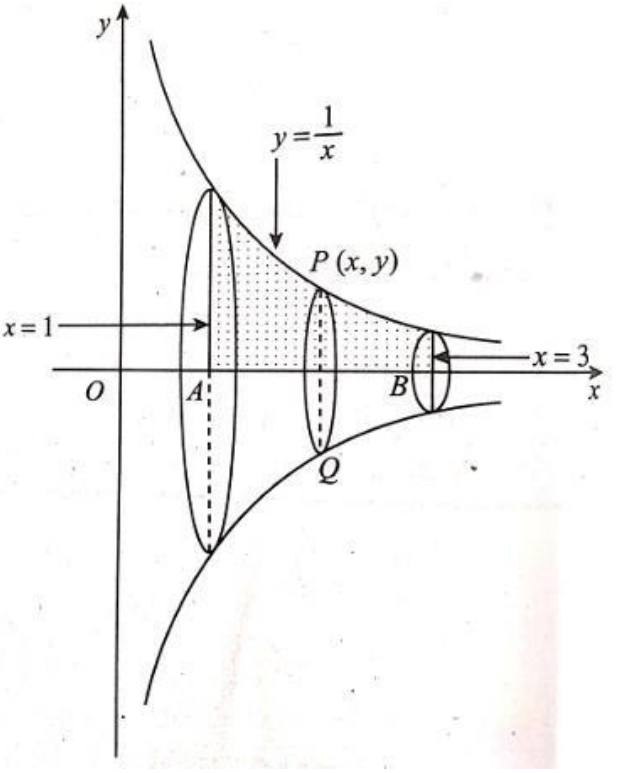
$$V = \int_a^b A(x)\, dx$$

$$= \int_1^3 \frac{\pi}{x^2}\, dx$$

$$= \pi \left(\frac{-1}{x}\right)_1^3$$

$$= \pi \left(\frac{-1}{3} + 1\right)$$

$$= \frac{2\pi}{3}$$



Example 3: Find the volume of the region between the curve $y = \sqrt{x}$, $0 \le x \le 4$ and the line x=4 revolved about X –axis to generate a solid

Solution:

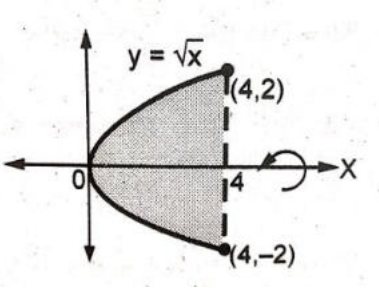
Here given that $y = \sqrt{x}$ $\quad 0 \le x \le 4$

Area of Cross- section,

$$A = \pi y^2$$

$$= \pi \cdot \left(\sqrt{x}\right)^2$$

$$= \pi \cdot x$$



The required volume is

$$V = \int_a^b A(x)\, dx$$

$$= \int_0^4 \pi\, x\, dx$$

$$= \pi \left(\frac{x^2}{2}\right)_0^4$$

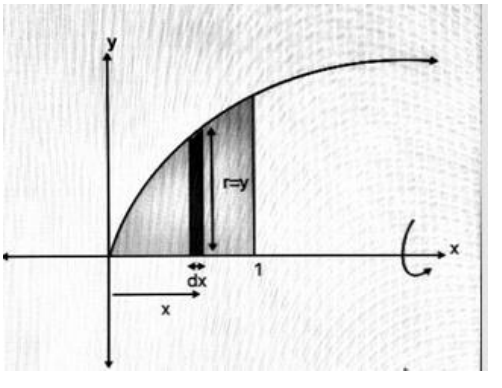
$$= \pi(8 - 0)$$

$$= 8\pi$$

Example 4: Using cylindrical shells, find the volume of the solid obtained by rotating about the x-axis the region under the curve $y = \sqrt{x}$ from 0 to 1

Solution:

Here $y = f(x) = \sqrt{x}$, $0 \le x \le 1$



The required volume by cylindrical shells is

$$V = 2\pi \int_a^b x\, f(x)\, dx$$

$$= 2\pi \int_0^1 x\sqrt{x}\, dx$$

$$= 2\pi \int_0^1 x^{\frac{3}{2}}\, dx$$

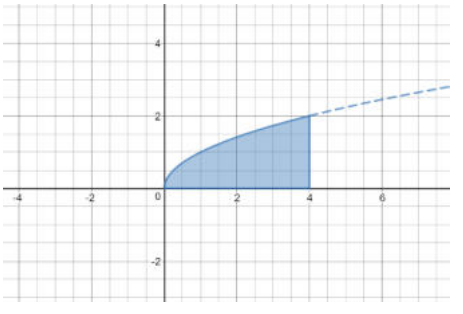
$$= 2\pi \left(\frac{x^{\frac{5}{2}}}{\frac{5}{2}}\right)_0^1$$

$$= \frac{4\pi}{5}(1 - 0) = \frac{4\pi}{5}$$

Example 5 : Find the volume of the region between the curve $y = \sqrt{x}$, $0 \le x \le 4$ and the line x=4 revolved about X –axis to generate a solid.

Solution:

The curve is $y = \sqrt{x}$,



The line $x = 4$ serves as the boundary in the x-direction.
We are revolving around the x -axis, so the solid is generated from $x = 0$ to $x = 4$

The required volume is

$$V = \int_a^b \pi\, y^2\, dx$$

$$= \int_0^4 \pi \left(\sqrt{x}\right)^2 dx$$

$$= \int_0^4 \pi\, x\, dx$$

$$= \pi \int_0^4 x\, dx$$

$$= \pi \left(\frac{x^2}{2}\right)_0^4$$

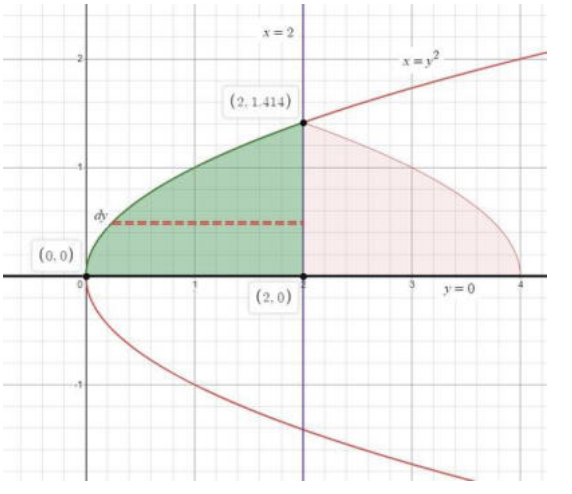
$$= \pi(8 - 0)$$

$$= 8\pi$$

Example 6 : Find the volume of solid generated by revolving the region bounded by x= y^2 and the lines x = 0, x= 2 about the x-axis.

Solution: Here the curve is x= y^2,

Area of Cross- section,



$$A = \pi y^2$$

$$= \pi \cdot x$$

The required volume is

$$V = \int_a^b A(x)\, dx$$

$$= \int_0^2 \pi\, x\, dx$$

$$= \pi \left(\frac{x^2}{2}\right)_0^2$$

$$= \pi(2 - 0)$$

$$= 2\pi$$

Example 7 : Find the volume of the solid obtained by rotating the region enclosed by the curves $y = x$ *and* $y = x^2$ about the x-axis.

Solution: First we find the point of intersection of $y = x$ *and* $y = x^2$

$$\therefore x = x^2$$

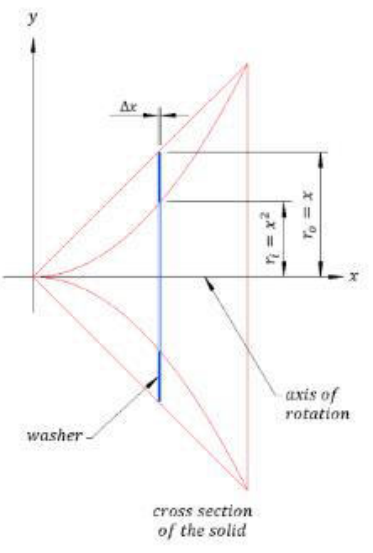
$$\therefore x^2 - x = 0$$

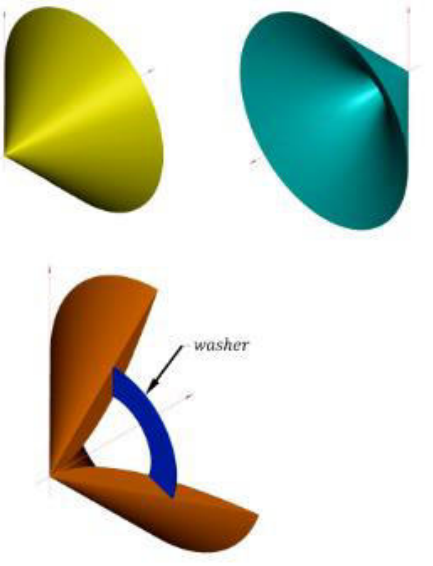
$$\therefore x(x - 1) = 0$$

So $x = 0$ and $x = 1$ are point of intersection.

The formula for the volume V of the solid of revolution using the washer method is:

$$V = \pi \int_a^b [R(x)^2 - r(x)^2]\, dx \quad \text{_____}(1)$$





Here:

- The outer radius $R(x) = x$
- The inner radius $r(x) = x^2$
- The bounds of integration are from $x = 0$ to $x = 1$

∴ from (1) the volume is

$$V = \pi \int_0^1 [(x)^2 - (x^2)^2]\, dx$$

$$= \pi \int_0^1 [x^2 - x^4]\, dx$$

$$= \pi \left(\frac{x^3}{3} - \frac{x^5}{5} \right)_0^1$$

$$= \pi \left(\frac{1}{3} - \frac{1}{5} \right)$$

$$= \frac{2\pi}{5}$$

Example 8 : Find the volume of solid generated by rotating the region bounded by $y = x^2 - 2x$ & $y = x$ about the lines $y = 4$.

Solution: First we find the point of intersection of the curves $y = x^2 - 2x$ & $y = x$

$$\therefore x = x^2 - 2x$$

$$\therefore x^2 - 2x - x = 0$$

$$\therefore x^2 - 3x = 0$$

$$\therefore x(x-3) = 0$$

So $x = 0$ and $x = 3$ are point of intersection.

The formula for the volume V of the solid of revolution using the washer method is:

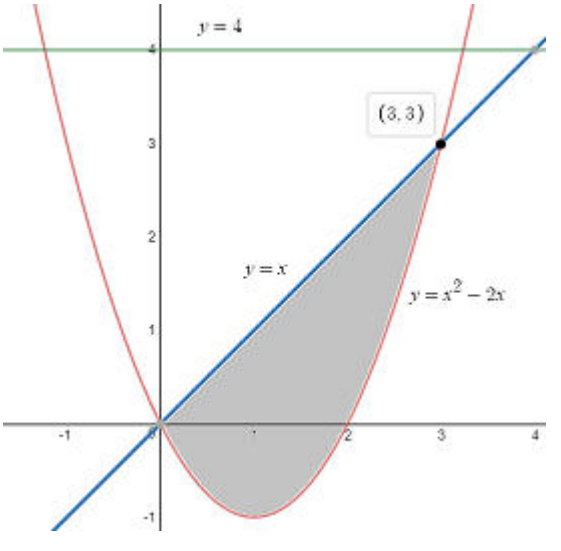
$$V = \pi \int_a^b [R(x)^2 - r(x)^2]\, dx _____(1)$$

- The outer radius $R(x)$ is the distance from the axis of rotation $y = 4$ to the curve that is farther from the axis.

  $\therefore\ R(x) = 4 - (x^2 - 2x)$

- The inner radius $r(x)$is the distance from the axis of rotation $y = 4$ to the curve that is closer to the axis.

  $\therefore r(x) = 4 - x$



From (1),

$$V = \pi \int_0^3 [(4 - (x^2 - 2x))^2 - (4 - x)^2]\, dx$$

$$= \pi \int_0^3 [(4 - x^2 + 2x)^2 - (x - 4)^2]\, dx$$

$$= \pi \int_0^3 [16 + x^4 + 4x^2 - 8x^2 - 4x^3 + 16x - x^2 + 8x - 16]\, dx$$

$$= \pi \int_0^3 [x^4 - 4x^3 - 5x^2 + 24x]\, dx$$

$$= \pi \left(\frac{x^5}{5} - x^4 - \frac{5x^3}{3} + 12x^2 \right)_0^3$$

$$= \pi \left(\frac{243}{5} - 81 - 45 + 108 \right)$$

$$= \pi \left(\frac{243}{5} - 18 \right)$$

$$= \left(\frac{153}{5} \right) \pi$$

Example 8 : Find the volume of the solid that results when the region enclosed by the curves $y = x^2$ and $x = y^2$ is revolved about the Y-axis.

Solution: Here the region is bounded by $y = x^2$ and $x = y^2$

we find the point of intersection

Substituting $y = x^2$ into $x = y^2$

$\therefore x = (x^2)^2$

$\therefore x = x^4$

$\therefore x^4 - x = 0$

$\therefore x(x^3 - 1) = 0$

$\therefore x = 0 \; or \; x = 1 \; \rightarrow y = 0 \; or \; y = 1$

So, the region is bounded by the points $(0,0) \; and \; (1,1)$

The required volume by cylindrical shells is

$$V = 2\pi \int_a^b x\, f(x)\, dx$$

$$= 2\pi \int_a^b x\left(y_2(x) - y_1(x)\right) dx$$

$$= 2\pi \int_0^1 x\left(\sqrt{x} - x^2\right) dx$$

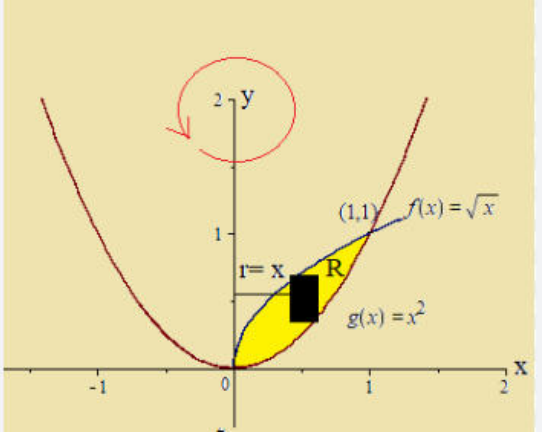
$$= 2\pi \int_0^1 \left(x^{\frac{3}{2}} - x^3\right) \; dx$$

$$= 2\pi \left(\frac{x^{\frac{5}{2}}}{5/2} - \frac{x^4}{4}\right)_0^1$$

$$= 2\pi \left(\frac{2}{5} - \frac{1}{4}\right)$$

$$= 2\pi \left(\frac{3}{20}\right)$$

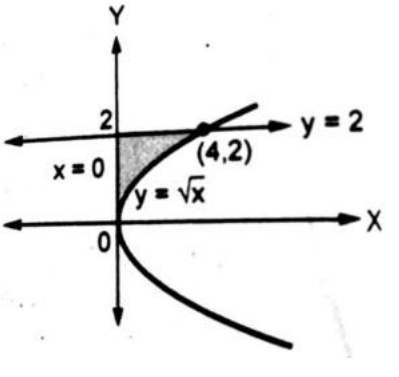
$$= \frac{3\pi}{10}$$



Example 9 : Find the volume of solid generated by revolving the region bounded by $y = \sqrt{x}$ and the lines $y = 2$ and $x = 0$ about the line (a) $x = 0$ *and* (b) y=2.

Solution: Given equation is $y = \sqrt{x}$ and $y = 2$, $x = 0$

(a) The required volume V generated by revolving shaded area about $x = 0$ Is given by



$$V = \pi \int_0^2 x^2 dy$$

$$= \pi \int_0^2 y^4 \, dy$$

$$= \pi \left(\frac{y^5}{5}\right)_0^2 = \frac{32\pi}{5}$$

(b) The required volume V generated by revolving shaded area about $x = 0$ Is given by

$$V = \pi \int_0^4 (2-y)^2 dx$$

$$= \pi \int_0^4 (2-\sqrt{x})^2 dx$$

$$= \pi \int_0^4 \left(4 - 4\sqrt{x} + x\right) dx$$

$$= \pi \left(4x - \frac{4\, x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{x^2}{2} \right)_0^4$$

$$= \pi \left(16 - \frac{64}{3} + 8 \right)$$

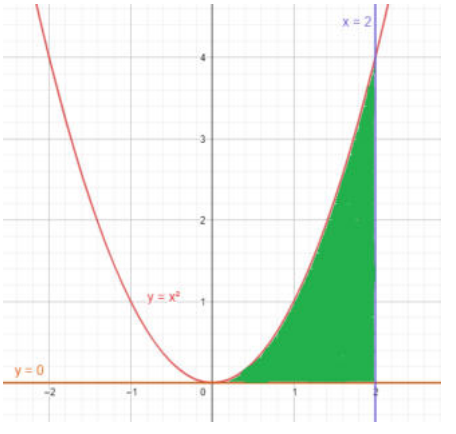
$$= \pi \left(\frac{48 - 64 + 8}{3} \right)$$

$$= \frac{8\pi}{5}$$

Example 10 : Find the volume of solid generated by revolving the region bounded by $y = x^2$, $y = 0$ and $x = 2$ about the $x - axis$.

Solution: Here the region is bounded by $y = x^2$, $y = 0$ and $x = 2$

The required volume is



$$V = \pi \int_a^b y^2\, dx$$

$$= \pi \int_0^2 (x^2)^2\, dx$$

$$= \pi \int_0^2 x^4\, dx$$

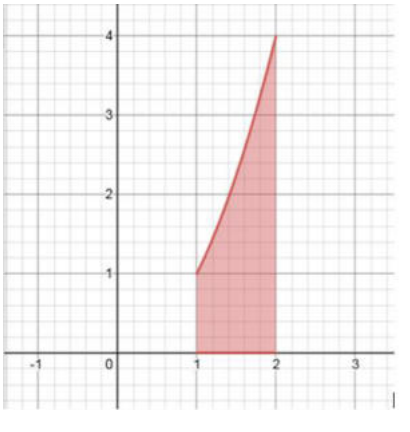
$$= \pi \left(\frac{x^5}{5}\right)_0^2$$

$$= \pi \left(\frac{32}{5}\right)$$

$$= \frac{32\pi}{5}$$

Example 11 : The graph of y = x^2 between x=1 and x=2 is rotated around the X-axis. Find the volume of a solid so generated

Solution:



The required volume is

$$V = \pi \int_a^b y^2\, dx$$

$$= \pi \int_1^2 (x^2)^2\, dx$$

$$= \pi \int_1^2 x^4\, dx$$

$$= \pi \left(\frac{x^5}{5}\right)_1^2$$

$$= \pi \left(\frac{32}{5} - \frac{1}{5}\right)$$

$$= \frac{31\pi}{5}$$

Example 12 : Find the volume generated by revolving the area bounded by

$2y = x^2, x = 4, y = 0$ about x –axis.

Solution: here the curve $2y = x^2$

$$\therefore y = \frac{x^2}{2}$$

The required volume is

$$V = \pi \int_a^b y^2 \, dx$$

$$= \pi \int_0^4 \left(\frac{x^2}{2}\right)^2 dx$$

$$= \frac{\pi}{4} \int_0^4 \frac{x^4}{4} dx$$

$$= \frac{\pi}{4}\left(\frac{256}{5}\right)$$

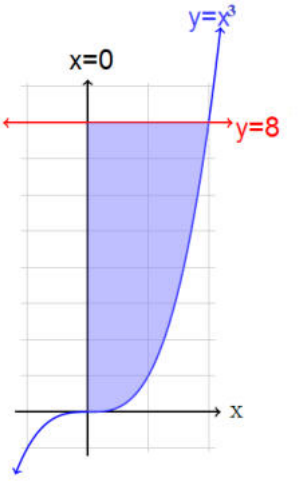
$$= \frac{64\,\pi}{5}$$

Example 13 : Find the volume of the solid obtained by rotating the region bounded by the curve $y = x^3$, $y = 8$ and $x = 0$ about the y-axis.

Solution:

Here the curve $y = x^3$

$$\therefore x = y^{\frac{1}{3}}$$



The required volume is

$$V = \pi \int_a^b x^2 \, dy$$

$$= \pi \int_0^8 \left(y^{\frac{1}{3}}\right)^2 dy$$

$$= \pi \int_0^8 y^{\frac{2}{3}} \, dy$$

$$= \pi \left(\frac{y^{\frac{5}{3}}}{5/3} \right)_0^8$$

$$= \frac{3\pi}{5} (8^{\frac{5}{3}})$$

$$= \frac{3\pi}{5} (2^5)$$

$$= \frac{3\pi}{5} (32)$$

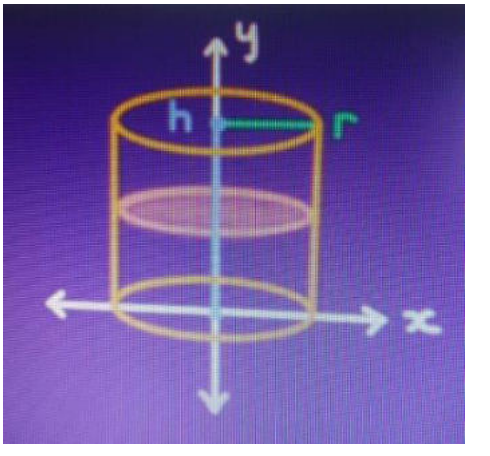
$$= \frac{96\,\pi}{5}$$

Example 13 : Using volume by slicing method, find the volume of a cylinder with radius 'r' and height 'h'

Solution: Slice the cylinder into thin circular disks of thickness dx perpendicular to its height h.

The radius of each disk is r (since the cylinder's radius is constant).

The area of each disk is $A = \pi r^2$



The required volume is

$$V = \int_a^b A(x)\, dx$$

$$= \int_0^h \pi r^2\, dx$$

$$= \pi r^2 \int_0^h 1\ dx$$

$$= \pi r^2\, [x]_0^h$$

$$= \pi r^2\, (h - 0)$$

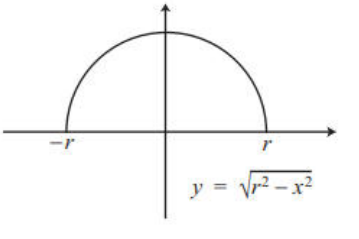
$$\therefore V = \pi r^2 h$$

Example 13 : Determine volume of a sphere of radius a by the solids of revolution

Solution: To determine the volume of a sphere using the method of solids of revolution, we can use the concept of rotating a curve around an axis. Specifically, we will rotate the upper half of the circle around the x-axis to generate the volume of a sphere.

- The equation $x^2 + y^2 = r^2$ represents the equation of a circle centred on the origin and with radius r. So the graph of the function $y = \sqrt{r^2 - x^2}$ is a semicircle.

We rotate this curve between x = −r and x = r about the x-axis



The required volume is

$$V = \int_a^b \pi y^2\, dx$$

$$= \int_{-r}^{r} \pi (r^2 - x^2)\, dx$$

$$= \pi \int_{-r}^{r} (r^2 - x^2)\, dx$$

$$= \pi \left(r^2 x - \frac{x^3}{3} \right)_{-r}^{r}$$

$$= \pi \left(r^3 - \frac{r^3}{3} + r^3 - \frac{r^3}{3} \right)$$

$$= 2\pi \left(r^3 - \frac{r^3}{3} \right)$$

$$= 2\pi \left(\frac{2r^3}{3} \right)$$

$$\therefore V = \frac{4}{3}\pi r^3$$

Example 14 : Use cylindrical shells to find volume of the solid obtained by rotating about the y-axis the region between $y = x$ and $y = x^2$.

Solution: Here the region is bounded by $y = x$ and $y = x^2$

we find the point of intersection

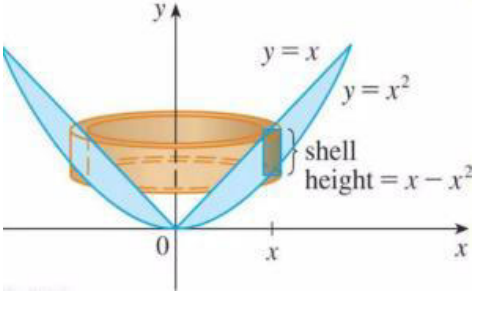
$\therefore x = x^2$

$\therefore x^2 - x = 0$

$\therefore x(x-1) = 0$

So $x = 0$ and $x = 1$

The formula for the volume using cylindrical shells when rotating about the y -axis is:



$$V = \int_a^b 2\pi x\, f(x)\, dx$$

$$= \int_a^b 2\pi x\, (y_1 - y_2) dx$$

$$= 2\pi \int_0^1 x\,(x - x^2)dx$$

$$= 2\pi \int_0^1 (x^2 - x^3)\,dx$$

$$= 2\pi \left(\frac{x^3}{3} - \frac{x^4}{4}\right)_0^1$$

$$= 2\pi \left(\frac{1}{3} - \frac{1}{4}\right)$$

$$= 2\pi \cdot \left(\frac{1}{12}\right)$$

$$= \frac{\pi}{6}$$

Example 15: Use the shell method to find volume of solid generated by revolving the region by $y = x,$ x=0 and y=1 about x-axis

Solution:

Volume of a solid of revolution about the x-axis is

$$V = \int_a^b 2\pi y\, f(y)\, dy$$

$$= \int_0^1 2\pi y\, y\, dy \qquad since\ f(y) = x = y$$

$$= 2\pi \int_0^1 y^2\ dy$$

$$= 2\pi \left(\frac{y^3}{3}\right)_0^1$$

$$= 2\pi \left(\frac{1}{3}\right)$$

$$= \frac{2\pi}{3}$$

Example 16: Find the volume of the solid of revolution of the area bounded by the curve $y = xe^x$ and the straight lines x=1, y=0.

Solution:

The required volume is

$$V = \pi \int_a^b y^2 \, dx$$

$$= \pi \int_0^1 (xe^x)^2 \, dx$$

$$= \pi \int_0^1 x^2 \, e^{2x} \, dx$$

$$= \pi \left[(x^2)\left(\frac{e^{2x}}{2}\right) - (2x)\left(\frac{e^{2x}}{4}\right) + (2)\left(\frac{e^{2x}}{8}\right) \right]_0^1$$

$$= \pi \left[\left(\frac{e^2}{2} - \frac{e^2}{2} + \frac{e^2}{4}\right) - \left(0 - 0 + \frac{1}{4}\right) \right]$$

$$= \pi \left[\left(\frac{e^2}{4}\right) - \left(\frac{1}{4}\right) \right]$$

$$= \frac{\pi}{4}(e^2 - 1)$$

❖ Area of Surface of Solid of Revolution (in Cartesian form)

- Area of surface generated by revolving the arc of the curve $y = f(x)$ about the x - axis is given by,

$$S = 2\pi \int_a^b y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

- Area of surface generated by revolving the arc of the curve $x = f(y)$ about the y - axis is given by,

$$S = 2\pi \int_c^d x \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dx$$

❖ Area of Surface of Solid of Revolution (in Polar form)

If $r = f(\theta)$ is the polar equation of the curve then the area of the surface of revolution generated by the curve for $\alpha \leq \theta \leq \beta$

(1) About the initial line $\theta = 0$ is

$$S = 2\pi \int_\alpha^\beta r \sin\theta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

(2) About the initial line $\theta = \frac{\pi}{2}$ is

$$S = 2\pi \int_\alpha^\beta r \cos\theta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

❖ Area of Surface of Solid of Revolution (in Parametric form)

- The area of surface generated by revolution the arc of the curve $x = f(t)$ and $y = g(t)$ from $t = a$ to $t = b$

(1) About the $x -$ axis is

$$S = 2\pi \int_{t=a}^{t=b} g(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

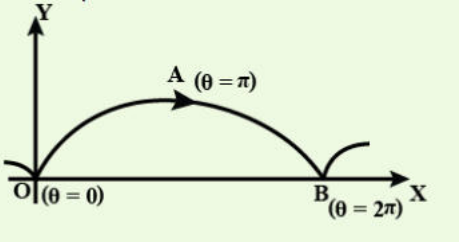
(2) About the $y -$ axis is

$$S = 2\pi \int_{t=a}^{t=b} f(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

Example 1: Find the surface area of the solid generated by revolving the cycloid $x = a(\theta - \sin\theta), y = a(1 - \cos\theta)$ About its base.

Solution:

The required surface area S is given by



$$S = 2\pi \int_0^{2\pi} y \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}\, d\theta$$

$$x = a(\theta - \sin\theta) \rightarrow \frac{dx}{d\theta} = a(1 - \cos\theta)$$

$$y = a(1 - \cos\theta) \rightarrow \frac{dy}{d\theta} = a \sin\theta$$

$$\therefore S = 2\pi \int_0^{2\pi} a(1 - \cos\theta) \sqrt{a^2(1 - \cos\theta)^2 + a^2 sin^2\theta}\, d\theta$$

$$= 2\pi \int_0^{2\pi} a(1 - \cos\theta)\ a\sqrt{1 - 2\cos\theta + cos^2\theta + sin^2\theta}\, d\theta$$

$$= 2\pi a^2 \int_0^{2\pi} (1 - \cos\theta) \sqrt{2 - 2\cos\theta}\, d\theta$$

$$= 2\pi a^2 \int_0^{2\pi} (1 - \cos\theta) \sqrt{2(1 - \cos\theta)}\, d\theta$$

$$= 2\pi a^2 \int_0^{2\pi} 2\, sin^2\frac{\theta}{2} \sqrt{2 \cdot 2\, sin^2\frac{\theta}{2}}\, d\theta$$

$$= 8\pi a^2 \int_0^{2\pi} sin^3\frac{\theta}{2}\, d\theta$$

Let $\frac{\theta}{2} = t \rightarrow d\theta = 2\,dt$ When $\theta = 0,$ $t = 0$ and When $\theta = 2\pi,$ $t = \pi$

$$= 8\pi a^2 \int_0^{\pi} sin^{\,3} t \cdot 2\,dt$$

$$= 16\pi a^2 \int_0^{\pi} sin^{\,3} t\;\; dt$$

$$= 32\pi a^2 \int_0^{\frac{\pi}{2}} sin^{\,3} t\;\; dt$$

$$= 32\pi a^2 \;\; \frac{2}{3 \times 1} \qquad\qquad by\ reduction\ formula$$

$$= \frac{64\,\pi a^2}{3}$$

Example 2: Find the area between the curve $y^2 = \frac{x^2}{1-x^2}$ and its asymptotes using improper integral**.**

Solution:

Here the give curve is $y^2 = \frac{x^2}{1-x^2}$

$$\therefore y = \pm\frac{x}{\sqrt{1-x^2}}$$

The asymptotes occur as x approaches ±1, where the denominator $\sqrt{1-x^2}$ tends to zero, making the expression unbounded.

The area between the curve and the x-axis from -1 to 1 can be found by integrating the positive branch $y = \frac{x}{\sqrt{1-x^2}}$ over this interval:

$$A = \int_{-1}^{1} y\,dx$$

$$A = 2\int_0^1 \frac{x}{\sqrt{1-x^2}}\,dx$$

Let $1 - x^2 = t \rightarrow -2xdx = dt$

$$\therefore 2xdx = -dt$$

$$A = -\int_1^0 \frac{dt}{\sqrt{t}}$$

$$= \int_0^1 \frac{dt}{\sqrt{t}}$$

$$= \int_0^1 t^{\frac{-1}{2}}\, dt$$

$$= \left(\frac{t^{1/2}}{1/2}\right)_0^1$$

$$= 2 \cdot (1 - 0)$$

$$= 2$$

Example 3 : Find the area of the surface of revolution of a quadrant of a circular arc as obtained by revolving it about a tangent at one of its ends

Solution: Let's assume the radius of the circular arc is R.

The equation of the circular arc in the first quadrant is

$$y = \sqrt{R^2 - x^2}$$

$$\therefore \frac{dy}{dx} = \frac{d}{dx}\left(\sqrt{R^2 - x^2}\right)$$

$$= \frac{1}{2\sqrt{R^2 - x^2}} \times (-2x)$$

$$\therefore \frac{dy}{dx} = \frac{-x}{\sqrt{R^2 - x^2}}$$

When this arc is revolved about the tangent at x=0, the surface area S is given by

$$S = 2\pi \int_a^b y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

$$= 2\pi \int_0^R y \sqrt{1 + \left(\frac{-x}{\sqrt{R^2 - x^2}}\right)^2}\, dx$$

$$= 2\pi \int_0^R y \sqrt{1 + \frac{x^2}{R^2 - x^2}}\, dx$$

$$= 2\pi \int_0^R y \sqrt{\frac{R^2 - x^2 + x^2}{R^2 - x^2}}\, dx$$

$$= 2\pi \int_0^R \sqrt{R^2 - x^2} \cdot \sqrt{\frac{R^2}{R^2 - x^2}}\, dx \qquad \because y = \sqrt{R^2 - x^2}$$

$$= 2\pi \int_0^R \sqrt{R^2 - x^2} \cdot \frac{R}{\sqrt{R^2 - x^2}}\, dx$$

$$= 2\pi \int_0^R R\, dx$$

$$= 2\pi R \int_0^R 1\, dx$$

$$= 2\pi R [x]_0^R$$

$$= 2\pi R^2$$